

Malignant Pericardial Effusion / Tamponade

- **Presentation:** Dyspnea, chest pain, tachycardia. Beck's Triad: **hypotension, jugular venous distention, muffled heart sounds. pulsus paradoxus.**
- **Management:**
 - **Unstable patients:** Emergent pericardiocentesis.
 - **Recurrent/malignant:** Pericardial window (surgical drainage).

Neuroendocrine Tumors (NETs)

- **Carcinoid syndrome:** **flushing, diarrhea, bronchospasm**—usually midgut primary with liver metastases.
- **Diagnosis:** 24-hour urine 5-HIAA (avoid bananas, pineapples, walnuts) + localization with DOTATATE PET/CT or octreotide scan.
- **Management:** Symptom control with somatostatin analogs (octreotide, lanreotide).

Febrile Neutropenia

- **Fever:** Single temperature $\geq 38.3^{\circ}\text{C}$ (101°F) OR sustained temperature $\geq 38.0^{\circ}\text{C}$ (100.4°F) for ≥ 1 hour.
- **Neutropenia:** Absolute Neutrophil Count (ANC) $< 500/\mu\text{L}$ or $< 1000/\mu\text{L}$ with predicted decline to $< 500/\mu\text{L}$.

Core Principles for Febrile Neutropenia Management:

1. This is a MEDICAL EMERGENCY.
2. Obtain blood cultures, then immediately initiate empiric broad-spectrum anti-pseudomonal antibiotics without delay.
3. **First-line antibiotics:** cefepime, piperacillin-tazobactam, or meropenem.
4. **Add vancomycin for:** hypotension, pneumonia, suspected catheter infection, or known MRSA colonization.

Drug Toxicities

- **Doxorubicin:** **Cardiomyopathy** (requires echocardiogram monitoring).
- **Bleomycin:** **Pulmonary Fibrosis.**
- **Cisplatin:** Nephrotoxicity and Ototoxicity.
- **Vincristine:** Peripheral **Neuropathy** (sensory and motor).
- **Cyclophosphamide:** **Hemorrhagic Cystitis** (prevent with Mesna).
- **Bevacizumab (VEGF inhibitor):** Hypertension (often reversible) and wound healing complications.
- **Immune checkpoint inhibitors (e.g., pembrolizumab):** unleash T-cells against self-antigens → autoimmune-like toxicity. Immune-related adverse events ("-itis" in any organ). Common: **colitis, pneumonitis, hepatitis, endocrinopathies.** Tx: hold therapy + high-dose corticosteroids.